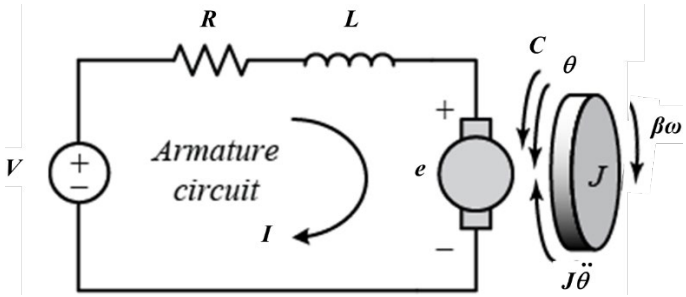
Course of Mechatronics - Exercitation 2: Response of a DC Motor

Consider the previous mechatronic system and concentrate on the mechanical part: the rotating masses. As a hypothesis, they are immersed in a viscous material. For this reason, the resistant torque can be modelled as a damping dissipating torque, depending on the rotational velocity ω of the rotating masses, according to the following expression:

$$C_R = \beta\omega$$

where β is the damping coefficient, measured in [*Nms/rad*].



The system must be studied for a step input of the developed torque at the rated condition (C=2.0 Ncm) with the same value of J (0.0000016 kgm^2).

The **objectives of the exercitation** are:

1. Find the required values of β to reach the steady-state condition, about ω, in a time equal to:
   i. 0.01 s
   ii. 0.025 s
   iii. 0.05 s
   iv. 0.075 s
   v. 0.1 s
   vi. 0.125 s
2. Find the steady-state values of ω for each value of the achieved β;
3. To study the behaviour of ω, solve the system according to the following:
   i. Direct Solution of the differential equation method;
   ii. State Space method;
   iii. Transfer Function method.

   implemented in Simulink;
4. Graphically, compare the results obtained by the previous methods;
5. Check if the time values of "*point 1.*" and the ω values of "*point 2.*" are satisfied.

Put all the contents of the present exercitation in a technical document to be delivered and discussed with the Lecturer at the end of the course. Students must prepare the technical paper as a result of the workgroup.